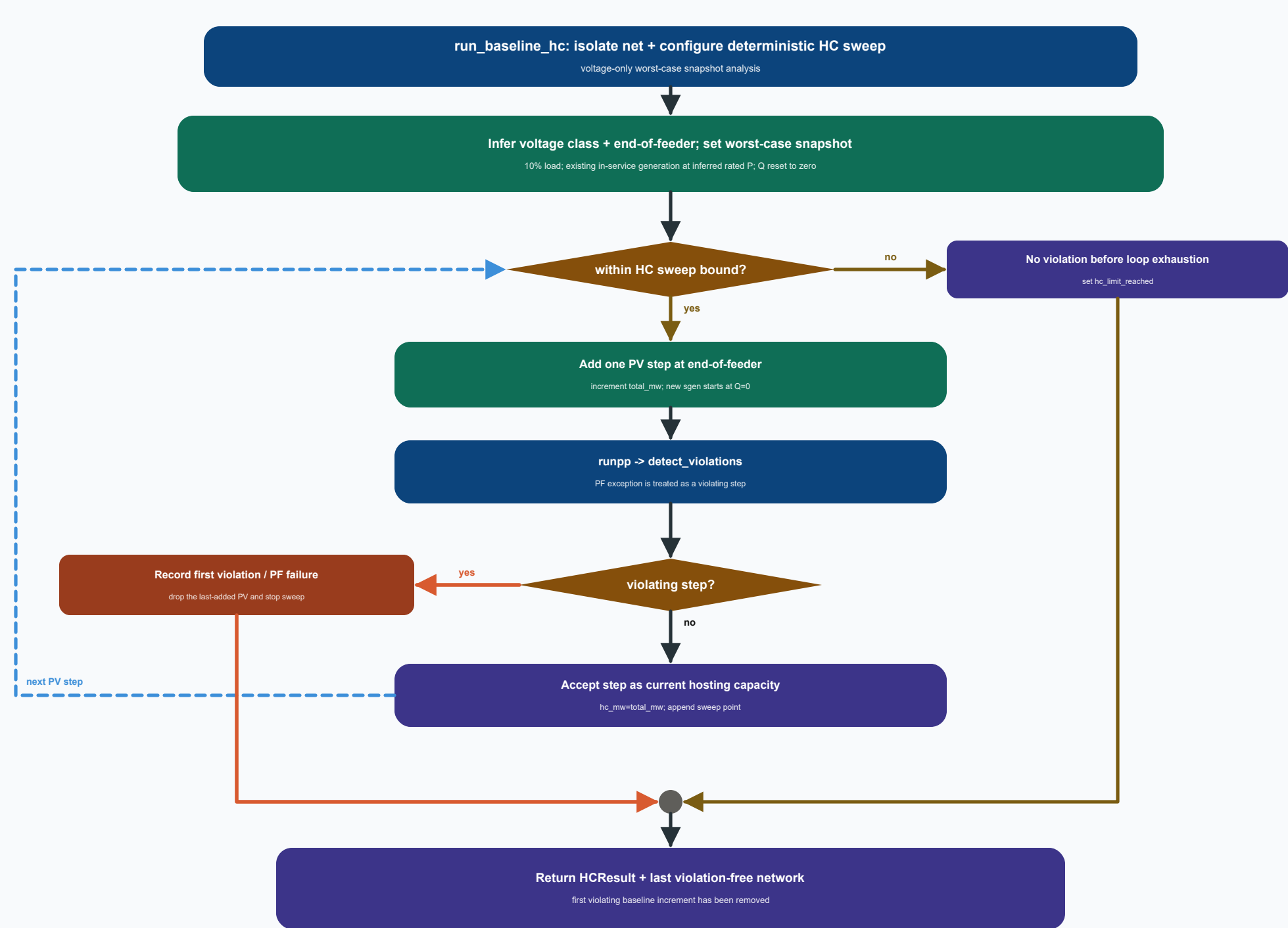




Deterministic HC setup

Both HC cases operate on a deep copy, so the network supplied by the benchmark runner is not modified.

The distribution level is inferred from the modal bus voltage: >1 kV selects MV parameters and <=1 kV selects LV parameters.

The end-of-feeder target is the reachable distribution-level bus with the greatest topological distance from the slack.

Before the sweep, load P/Q is reduced to 10% and existing in-service generators are placed at inferred rated active power.

The analysis evaluates voltage violations only; every power flow forces voltage-dependent loads off and uses Newton-Raphson.

Incremental baseline sweep

Each iteration adds one PV sgen at the same end-of-feeder bus and increases total_mw by the MV or LV step size.

A solved snapshot is passed to detect_violations; a power-flow exception is handled as a violating step.

The first violating step records the binding voltage location and is removed from the returned baseline network.

Every clean step becomes the latest hosting capacity and contributes one {mw, max_vm_pu} point to sweep_curve.

If the loop finishes without a violating step, hc_limit_reached is set and the last accepted capacity is retained.

Result semantics

HCResult stores hc_mw, the first violating MW level, binding bus/voltage, step count, sweep limit status and sweep curve.

Baseline HC additionally returns the deep-copied network at the last accepted PV capacity for downstream stressed analysis.

MV uses 0.5 MW increments up to the configured 20 MW bound; LV uses 0.01 MW increments up to 0.5 MW.

No Monte Carlo placement or annual QSTS is performed here; this module implements the deterministic snapshot method.